FEAT-ZD-48600 · DS-06 Unit Test Cases · Unit tests and coverage (≥90%)

Data Science agent · item task.unit_tests · source hobasa-data-science-agent/outputs/FEAT-ZD-48600/task.unit_tests.md · approved by automated checks and independent judge

task.unit_tests: FEAT-ZD-48600

Unit tests for the Data Science lane of FEAT-ZD-48600. This is step DS-06, Unit Test Cases. The workbook outcome is "Unit testing completed". The tests cover the code that DS-05 changed in `ap_forecast` on branch `feature/story`: STORY-03 (`vendor_risk`) and STORY-04 (`risk=None` on `build_explanation_request`). The tests are not committed yet. The spine commits them after evaluation.

Field	Value	Source
Trace ID	FEAT-ZD-48600	record:brief
Step	DS-06 Unit Test Cases	record:brief
Owner / gate	Developer / no gate: automated checks, the independent judge and sampled human evaluation	record:brief
Inputs	task.code (7b73e9c), feat.stories (0590ea4)	record:brief
Repository / branch	`ap_forecast`, `feature/story`	record:brief; task.code
Allowed paths in this step	`tests/`	record:brief
CI command	`/usr/bin/python3 ci.py` (coverage gate `COVERAGE_MIN = 90.0`)	record:brief; file:ap_forecast/ci.py

1. Result

CI passes: every test is green and line coverage of `src/ap_forecast` is 100.0%, above the 90.0 gate.

CI check	Result	Source
Compile `src`	`true`	tool:ci (`/usr/bin/python3 ci.py`)
Tests run / failures / errors	25 / 0 / 0	tool:ci
Coverage of `src/ap_forecast`	100.0 (32 statements, 0 missed)	tool:ci
Static issues	none (`[]`)	tool:ci
`passed`	`true`, exit code 0	tool:ci

Per-module coverage from the CI report:

Module	Statements	Missed	Cover
`src/ap_forecast/__init__.py`	5	0	100%
`src/ap_forecast/explain.py`	11	0	100%
`src/ap_forecast/predict.py`	5	0	100%
`src/ap_forecast/risk.py`	11	0	100%

The coverage comes from the tests alone. There are no exclusion pragmas, `ci.py` is not edited, and no source file is changed in this step (tool:repo; `git status`).

2. Files changed

File	Change	Story
`tests/test_risk.py`	New: 15 tests in `VendorRiskThresholdTest`, `VendorRiskIsolationTest` and `VendorRiskExportTest`	STORY-03
`tests/test_explain_risk.py`	New: 7 tests in `ExplanationRiskTest`	STORY-04
`tests/test_forecast.py`	Unchanged. Its 3 existing tests still pass, so the existing call forms still work (AC-04.3, NFR-4)	—

feat.stories SA-6 lets a lane add new test files instead of extending the existing one. Using new files leaves the existing tests exactly as they were.

Blast radius from CodeGraph (mcp:codegraph): `tests/test_risk.py` covers `vendor_risk`. `tests/test_forecast.py` and `tests/test_explain_risk.py` cover `build_explanation_request`.

3. Acceptance criteria to tests

3.1 STORY-03: `vendor_risk(history)` *(*`tests/test_risk.py`*)*

AC	Test	Input → expected
AC-03.1	`test_above_30_is_high`	`[31]` → `'high'`; `[45, 60]` → `'high'`
AC-03.2	`test_exactly_30_is_medium`	`[30]` → `'medium'` (boundary day, L-007)
AC-03.3	`test_between_10_and_30_is_medium`	`[11]`, `[20, 25]` → `'medium'`
AC-03.4	`test_exactly_10_is_low_and_11_is_medium`	`[10]` → `'low'`; `[11]` → `'medium'` (boundary day, L-007)
AC-03.5 [Pending PQ-1]	`test_empty_history_is_low`	`[]` → `'low'`
AC-03.6 [Pending TQ-5]	`test_early_payments_only_are_low`	`[-5, -20]` → `'low'`
AC-03.6 / predict rule	`test_negative_delays_count_as_zero_not_netted`	`[-100, 62]` predicts 31, so `'high'`: a negative delay counts as 0 and is not netted against late payments
AC-03 preamble (rounded prediction)	`test_threshold_applies_to_rounded_prediction`	`[30, 30, 31]` → `'medium'`; `[10, 10, 11]` → `'low'`; `[31, 31, 30]` → `'high'`
C-1 "predict once" (task.code)	`test_generator_history`	a generator of `[31, 31]` → `'high'`
C-1 named constants (task.code)	`test_constants`	`HIGH_ABOVE_DAYS == 30`, `MEDIUM_ABOVE_DAYS == 10`
AC-03.7	`test_no_config_read_or_llm_request`	`explain.load_config` and `explain.build_explanation_request` are patched to raise; `vendor_risk` runs on five histories and calls neither
AC-03.7	`test_module_imports_only_typing_and_predict`	the AST of `risk.py` imports exactly `typing` and `.predict`, with no `explain`, network or config import
AC-03.8	`test_exported_in_all`	`'vendor_risk'` is in `ap_forecast.__all__`, and the package attribute is `risk.vendor_risk`
NFR-4	`test_existing_exports_kept`	`predict_delay_days` and `build_explanation_request` stay in `__all__`
AC-03.8	`test_returns_only_known_levels`	single-value histories from -5 to 60 give exactly `{'high', 'medium', 'low'}`

3.2 STORY-04: `build_explanation_request(..., risk=None)` *(*`tests/test_explain_risk.py`*)*

AC	Test	Input → expected
AC-04.1	`test_risk_appended_to_prompt`	for `'high'`, `'medium'` and `'low'`, the prompt is today's prompt followed by "`Risk level: <risk>.`"
AC-04.1	`test_risk_positional_fourth_argument`	`risk` passed as the 4th positional argument gives the same result
AC-04.2	`test_without_risk_request_is_unchanged`	with no `risk`, the whole returned dict equals today's dict
AC-04.5 [Pending TQ-4]	`test_explicit_none_is_not_given`	`risk=None`, by keyword or by position, gives the same request as no `risk`, with no "Risk level" text
AC-04.3	`test_existing_positional_config_call`	`config` passed positionally still binds to `config`
AC-04.4	`test_provider_model_tokens_from_config_with_risk`	with `risk`, provider, model and `max_tokens` come from the explicit config, and there is one user message
AC-04.4	`test_provider_model_tokens_from_models_json_with_risk`	with no config, provider, model and `max_tokens` equal `load_config()` (`models.json`)

The test data uses only the synthetic vendor name "Synthetic Vendor A". The tests contain no real vendor, no secret and no personal data.

4. Not tested, on purpose

Item	Why	Source
`__version__` value `0.10.0`	It is still open as DSQ-1 (Developer). A test would pin an unconfirmed value	task.code (section 7)
A `risk` value other than the three labels, such as `""` or `'unknown'`	AC-04.1 appends any value that is not None, but what counts as "given" is still pending TQ-4. The tests use only the three labels `vendor_risk` returns	feat.stories (AC-04.5, TQ-4)
Sending the request to `acme-llm`	The code only builds the request and does not send it, so there is no network behaviour to test. Whether the label may go to the provider is TQ-8	task.code (sections 2.2, 6); feat.stories (STORY-04 open items)

5. Secrets and environments

The tests need no secret and no network. They read only files inside the repository: the packaged `models.json` (through `load_config`, in `tests/test_explain_risk.py` and `tests/test_forecast.py`) and the source of `src/ap_forecast/risk.py` (through `Path(risk_module.__file__).read_text()` in `test_module_imports_only_typing_and_predict`). Per-environment confirmation of secrets access is still the non-blocking SSQ-2 carried by task.code. Nothing in this step depends on it.

6. Notes

- The full test suite runs through the CI, so the tests that cover each changed symbol (section 2) all run.
- Python's `round` rounds halves to even, so a history whose mean is exactly 30.5 predicts 30 and gives `'medium'`. This is how the existing `predict_delay_days` behaves, and this step does not change it.
- Two test histories have means that end in exactly .5: `[45, 60]` (mean 52.5, predicts 52) and `[20, 25]` (mean 22.5, predicts 22). Both are far from the thresholds, so rounding up or down gives the same level (`'high'` and

`'medium'`). The cases that test rounding at a threshold, in `test_threshold_applies_to_rounded_prediction`, use means of about 30.33, 10.33 and 30.67, none of which ends in .5.

7. Open questions (carried, none blocking)

ID	Question	Owner
DSQ-1	Confirm `0.10.0` as the minor version step from `0.9.0`	Developer
DSQ-2	The coverage gate is met in DS-06 (100.0 ≥ 90.0); please confirm this closes DSQ-2	Developer
PQ-1, TQ-4, TQ-5	The tests for AC-03.5, AC-03.6 and AC-04.5 follow the interim design. If an answer changes that design, only those tests change	PM / Architect
TQ-8, PQ-5	Whether the risk label may be sent to `acme-llm`; the order of this change relative to PROV-ACME-2026-11	Architect / PM

Tool results (recorded by the spine from the real tools)

[Tool result] [ap_forecast] Unit tests: 25 run, 0 failing; coverage 100.0%